Motor C: Forecast Dietario con Prophet (2023–2030)

Aguacate Team · TFM La Dieta de un País

Este notebook entrena modelos Prophet por país × macrocategoría y genera proyecciones de la composición dietaria hasta 2030.
El vector dietario proyectado se pasa al LightGBM guardado (`lgb_final.pkl`) para obtener el CO_2 estimado futuro.

Limitación documentada: Prophet se entrena con 13 observaciones anuales (2010-2022).
Los intervalos de confianza al 80% son razonables hasta 2025-2026 pero se amplían considerablemente hacia 2030.
Interpretar como *escenario tendencial basado en inercia histórica*, no como predicción exacta.

In [1]:
```
import pandas as pd
import numpy as np
import joblib
import warnings
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
warnings.filterwarnings('ignore')

from prophet import Prophet

SEED = 42
np.random.seed(SEED)

MACROS = [
    'pct_Cereales', 'pct_Tuberculos', 'pct_Azucares',
    'pct_Aceites_Grasas', 'pct_Carnes', 'pct_Lacteos_Huevos', 'pct_Frutas_Verduras'
]
FEATURES_LGB = MACROS + ['log_Food_CPI', 'cluster_id']
HORIZON_YEAR = 2030
HORIZON_PERIODS = HORIZON_YEAR - 2022  # 8 años

MACRO_COLORS = {
    'pct_Cereales':       '#f0a500',
    'pct_Tuberculos':     '#8b4513',
    'pct_Azucares':       '#e84393',
    'pct_Aceites_Grasas': '#ffd700',
    'pct_Carnes':         '#c0392b',
    'pct_Lacteos_Huevos': '#a0a0a0',
    'pct_Frutas_Verduras':'#2ecc71'
}

print('Librerías cargadas')
```

```
Librerías cargadas
```

In [2]:
```
# ── CARGA DE DATOS Y MODELO ──────────────────────────────────────
df = pd.read_parquet('../processed/03_clusters.parquet')
lgb_model = joblib.load('../processed/lgb_final.pkl')

print(f'Dataset: {df.shape}')
```

```
print(f'Países: {df["Area"].nunique()} | Años: {df["Year"].min()}–{df["Year"].max()}'
print(f'Modelo LightGBM cargado: {type(lgb_model).__name__}')
```

```
Dataset: (390, 15)
Países: 30 | Años: 2010–2022
Modelo LightGBM cargado: LGBMRegressor
```

In [3]:

```
# ── FUNCIÓN PROPHET POR SERIE ──────────────────────────────────────
def forecast_serie(series_df, col, country, horizon=HORIZON_PERIODS):
    """
    Entrena Prophet sobre 13 puntos anuales y genera forecast.
    Parámetros conservadores para series cortas:
      - changepoint_prior_scale=0.1 (evita overfitting)
      - seasonality desactivada (datos anuales, no hay ciclos intrayear)
    """
    prophet_df = series_df[['Year', col]].copy()
    prophet_df.columns = ['ds', 'y']
    prophet_df['ds'] = pd.to_datetime(prophet_df['ds'].astype(str) + '-01-01')

    m = Prophet(
        yearly_seasonality=False,
        weekly_seasonality=False,
        daily_seasonality=False,
        seasonality_mode='additive',
        changepoint_prior_scale=0.1,
        interval_width=0.80
    )
    m.fit(prophet_df)

    future = m.make_future_dataframe(periods=horizon, freq='YS')
    forecast = m.predict(future)
    forecast['Area'] = country
    forecast['variable'] = col

    # Clamp proporciones entre 0 y 1
    for c in ['yhat', 'yhat_lower', 'yhat_upper']:
        forecast[c] = forecast[c].clip(0, 1)

    return forecast[['Area', 'variable', 'ds', 'yhat', 'yhat_lower', 'yhat_upper']]

print('Función forecast_serie definida')
```

```
Función forecast_serie definida
```

In [4]:

```
# ── BUCLE: 30 países × 7 macrocategorías = 210 modelos Prophet ──
records = []
countries = sorted(df['Area'].unique())

for i, country in enumerate(countries):
    sub = df[df['Area'] == country].sort_values('Year')
    for macro in MACROS:
        fc = forecast_serie(sub, macro, country)
        records.append(fc)
    if (i + 1) % 10 == 0:
        print(f'  {i+1}/{len(countries)} países procesados...')

forecast_macros = pd.concat(records, ignore_index=True)
forecast_macros['Year'] = forecast_macros['ds'].dt.year

print(f'\nForecast macros shape: {forecast_macros.shape}')
print(f'Años cubiertos: {forecast_macros["Year"].min()}–{forecast_macros["Year"].max(
print(f'Países: {forecast_macros["Area"].nunique()}')
```

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10/30 países procesados...

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20/30 países procesados...
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  30/30 países procesados...

Forecast macros shape: (4410, 7)
Años cubiertos: 2010–2030
Países: 30
```

In [5]:

```
# ── CONSTRUIR VECTORES DIETARIOS Y PREDECIR CO₂ ──────────────────

# Pivotar: una fila por (país, año), columnas = macrocategorías
pivot = forecast_macros.pivot_table(
    index=['Area', 'Year'],
    columns='variable',
    values='yhat'
).reset_index()
pivot.columns.name = None

# cluster_id: usar el último cluster conocido por país (98.7% estabilidad temporal)
last_cluster = df.groupby('Area')['cluster_id'].last().reset_index()
pivot = pivot.merge(last_cluster, on='Area', how='left')

# log_Food_CPI: último valor conocido como proxy constante
# Limitación documentada: no se modela la evolución del CPI
last_cpi = df.groupby('Area')['Food_CPI'].last().reset_index()
last_cpi['log_Food_CPI'] = np.log(last_cpi['Food_CPI'])
pivot = pivot.merge(last_cpi[['Area', 'log_Food_CPI']], on='Area', how='left')

# Re-normalizar proporciones a que sumen 1 (constraint composicional)
row_sums = pivot[MACROS].sum(axis=1)
pivot[MACROS] = pivot[MACROS].div(row_sums, axis=0)

# Predecir CO₂ con LightGBM
pivot['CO2eq_forecast'] = lgb_model.predict(pivot[FEATURES_LGB]).astype('float32')

print(f'Pivot shape: {pivot.shape}')
print(f'CO2 proyectado — rango: {pivot["CO2eq_forecast"].min():.2f} — {pivot["CO2eq_f
```

```
Pivot shape: (630, 12)
CO2 proyectado — rango: 0.73 — 11.41 t/cápita
```

In [6]:

```
# ── GUARDAR PARQUET ──────────────────────────────────────────────
co2_fc = pivot[['Area', 'Year', 'CO2eq_forecast']]
output = forecast_macros.merge(co2_fc, on=['Area', 'Year'], how='left')

output.to_parquet('../processed/05_forecast_dietario.parquet', index=False)
print(f'Guardado: processed/05_forecast_dietario.parquet')
print(f'Shape: {output.shape}')
print(f'Columnas: {output.columns.tolist()}')
```

```
Guardado: processed/05_forecast_dietario.parquet
Shape: (4410, 8)
Columnas: ['Area', 'variable', 'ds', 'yhat', 'yhat_lower', 'yhat_upper', 'Year', 'CO2e
q_forecast']
```

In [7]:

```
# ── VALIDACIÓN VISUAL: 3 países representativos ──────────────────
# Spain (C0-Proteica), India (C2-Cereal), Nigeria (C1-Tuberosa)
VALID_COUNTRIES = ['Spain', 'India', 'Nigeria']

fig, axes = plt.subplots(len(VALID_COUNTRIES), 2, figsize=(16, 5 * len(VALID_COUNTRIE
```

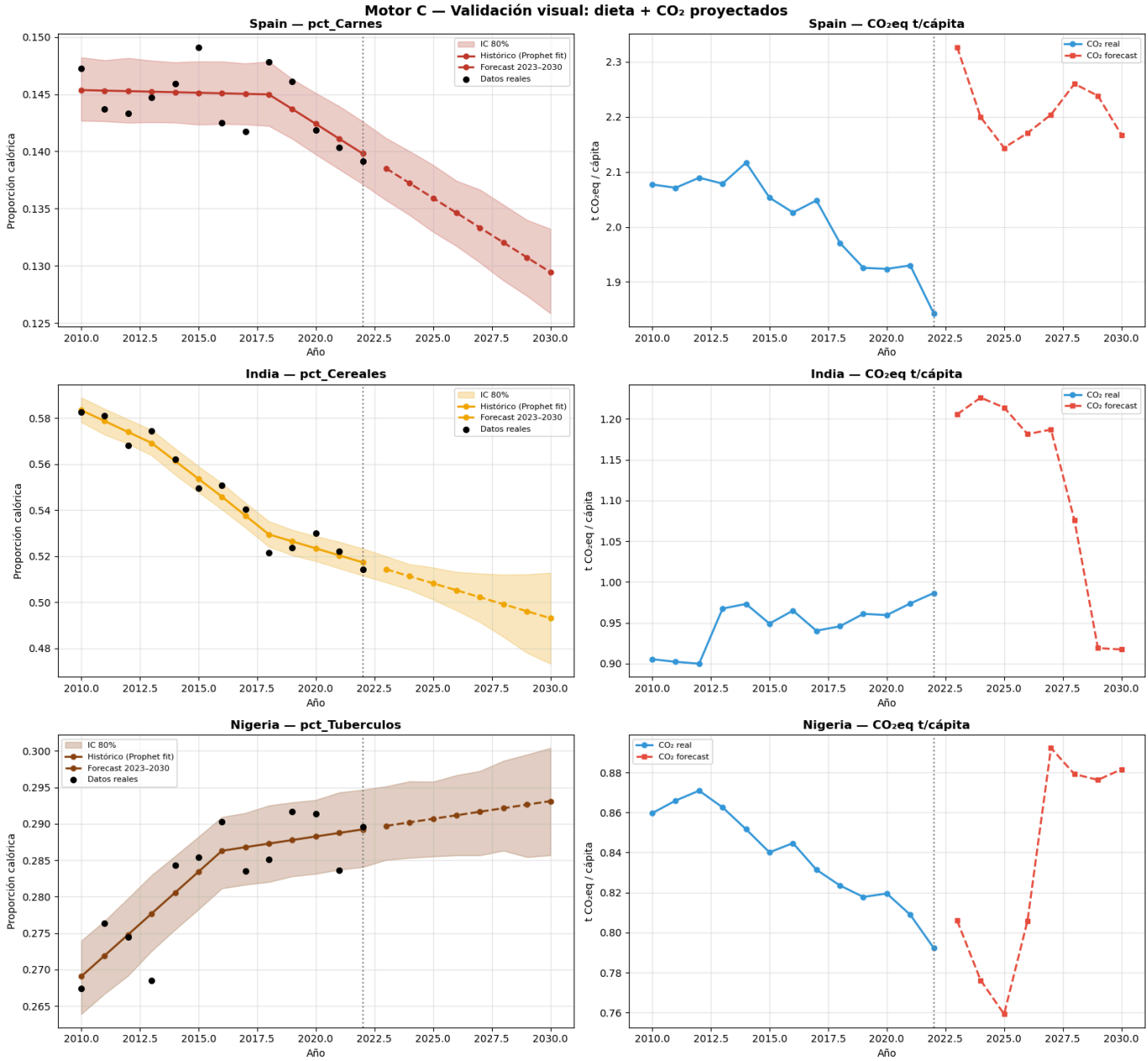
```
for row, country in enumerate(VALID_COUNTRIES):
    # ── Dieta histórica + forecast para 1 macrocategoría representativa ──
    macro_plot = 'pct_Carnes' if country == 'Spain' else 'pct_Cereales' if country ==
    fc_country = output[(output['Area'] == country) & (output['variable'] == macro_pl
    hist_country = df[df['Area'] == country].sort_values('Year')

    ax = axes[row, 0]
    ax.fill_between(
        fc_country['Year'],
        fc_country['yhat_lower'],
        fc_country['yhat_upper'],
        alpha=0.25, color=MACRO_COLORS[macro_plot], label='IC 80%'
    )
    ax.plot(fc_country[fc_country['Year'] <= 2022]['Year'],
            fc_country[fc_country['Year'] <= 2022]['yhat'],
            '-o', color=MACRO_COLORS[macro_plot], lw=2, ms=5, label='Histórico (Proph
    ax.plot(fc_country[fc_country['Year'] > 2022]['Year'],
            fc_country[fc_country['Year'] > 2022]['yhat'],
            '--o', color=MACRO_COLORS[macro_plot], lw=2, ms=5, label='Forecast 2023–2
    ax.scatter(hist_country['Year'], hist_country[macro_plot],
               color='black', zorder=5, s=30, label='Datos reales')
    ax.axvline(2022, color='gray', linestyle=':', lw=1.5)
    ax.set_title(f'{country} — {macro_plot}', fontweight='bold')
    ax.set_xlabel('Año'); ax.set_ylabel('Proporción calórica')
    ax.legend(fontsize=8); ax.grid(alpha=0.3)

    # ── CO₂ histórico + forecast ──
    co2_hist = hist_country[['Year', 'CO2eq_t_per_capita']]
    co2_proj = pivot[(pivot['Area'] == country)][['Year', 'CO2eq_forecast']]

    ax2 = axes[row, 1]
    ax2.plot(co2_hist['Year'], co2_hist['CO2eq_t_per_capita'],
             '-o', color='#3498db', lw=2, ms=5, label='CO₂ real')
    ax2.plot(co2_proj[co2_proj['Year'] > 2022]['Year'],
             co2_proj[co2_proj['Year'] > 2022]['CO2eq_forecast'],
             '--s', color='#e74c3c', lw=2, ms=5, label='CO₂ forecast')
    ax2.axvline(2022, color='gray', linestyle=':', lw=1.5)
    ax2.set_title(f'{country} — CO₂eq t/cápita', fontweight='bold')
    ax2.set_xlabel('Año'); ax2.set_ylabel('t CO₂eq / cápita')
    ax2.legend(fontsize=8); ax2.grid(alpha=0.3)

plt.suptitle('Motor C — Validación visual: dieta + CO₂ proyectados',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('../processed/05_forecast_validacion.png', dpi=150, bbox_inches='tight')
plt.show()
print('Guardado: processed/05_forecast_validacion.png')
```




```
Guardado: processed/05_forecast_validacion.png
```

In [8]:

```python
# ── TABLA RESUMEN: CO₂ 2022 vs proyectado 2030 ──────────────────
co2_2022 = df[df['Year'] == 2022][['Area', 'CO2eq_t_per_capita']].rename(
    columns={'CO2eq_t_per_capita': 'CO2_2022'})
co2_2030 = pivot[pivot['Year'] == 2030][['Area', 'CO2eq_forecast']].rename(
    columns={'CO2eq_forecast': 'CO2_2030'})

resumen = co2_2022.merge(co2_2030, on='Area')
resumen['Delta_pct'] = ((resumen['CO2_2030'] - resumen['CO2_2022']) / resumen['CO2_20
resumen = resumen.sort_values('Delta_pct')

print('CO₂ proyectado 2030 vs observado 2022 (variación %):')
print(resumen.to_string(index=False))
```

```
CO₂ proyectado 2030 vs observado 2022 (variación %):
                                                  Area  CO2_2022  CO2_2030  Delta_pct
                                                Canada  7.434406  2.385381 -67.900002
                                                Brazil  9.748733  3.195487 -67.199997
                                              Thailand  3.268874  2.287697 -30.000000
                                                Poland  3.586438  2.992711 -16.600000
                                               Romania  2.986566  2.540742 -14.900000
                                              Ethiopia  1.217610  1.073473 -11.800000
                                                 India  0.986313  0.917199  -7.000000
                                                Morocco  1.266582  1.181276  -6.700000
                                             Australia  7.687373  7.266189  -5.500000
                                           Saudi Arabia  2.354669  2.312242  -1.800000
                                               Colombia  3.326870  3.423951   2.900000
United Kingdom of Great Britain and Northern Ireland  1.858086  1.951841   5.000000
                                            Kazakhstan  2.779733  3.042281   9.400000
                                               Nigeria  0.792288  0.881542  11.300000
                                             Argentina  5.687041  6.397759  12.500000
                                            Bangladesh  0.847154  0.959786  13.300000
                                                Mexico  1.911802  2.188830  14.500000
                                                 Spain  1.842866  2.167403  17.600000
                                                France  2.133915  2.515600  17.900000
                                               Germany  2.096628  2.560986  22.100000
                                                 Italy  1.746279  2.297462  31.600000
                                 United States of America  3.028903  4.055270  33.900002
                                               Senegal  1.184145  1.623401  37.099998
                                                 Egypt  0.777203  1.157766  49.000000
                                              Pakistan  1.163849  1.800860  54.700001
                                        China, mainland  1.586632  2.484232  56.599998
                                             Indonesia  2.042389  3.580914  75.300003
                                                 Kenya  1.059605  2.968747 180.199997
                                        Republic of Korea  1.895967  5.458271 187.899994
                                              Viet Nam  1.440029  4.173141 189.800003
```

Resumen ejecutivo — Motor C

- **210 modelos Prophet** entrenados (30 países × 7 macrocategorías)
- **Horizonte:** 2023–2030 (8 años de forecast)
- **CO_2 proyectado:** derivado del vector dietario proyectado aplicado al `lgb_final.pkl`
- **Limitación principal:** series de 13 observaciones anuales → intervalos amplios a partir de 2026
- **Output:** `processed/05_forecast_dietario.parquet`
- **Uso en dashboard:** página `05_forecast_dietario.py` del simulador Streamlit